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Ontogeny influences tree growth response to soil fertility and neighbourhood crowding in an old-growth temperate forest

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- **Background and aims:** Abiotic and biotic factors simultaneously affect tree growth and thus shape community structure and dynamics. In particular, trees of different size classes show different growth responses to soil nutrients and neighbourhood crowding, but our understanding of how species' joint responses to these factors vary between size classes remains limited in multi-storied temperate forests. Here, we investigated size class differences in tree growth response to soil gradients and neighbourhood crowding in an old-growth temperate forest.
- **Methods:** We combined growth data over 15 years from 38 902 individuals of 42 tree species with trait data in a 25-ha temperate forest plot in northeast China. We built hierarchical Bayesian models of tree growth to examine the effects of soil gradients and neighbourhood crowding between size classes and canopy types.
- **Key results:** We found that soil and neighbours mainly acted separately in shaping tree growth in small and large trees. Soil total nitrogen and phosphorus increased tree growth in small trees, in particular of understorey species, but not in large trees. Neighbours reduced tree growth in both tree size classes, with stronger effects on large than small trees, and on canopy than understorey species. Furthermore, small trees with higher specific leaf area grew faster in fertile soils, and small trees with less seed mass grew faster in crowded environments. Large trees with higher specific leaf area, specific root length and less seed mass grew faster in crowded environments, while these traits had limited influence on tree growth response to soil gradients.
- **Conclusions:** Our study highlights the importance of size class in modulating the response of tree growth to soil and neighbours, and the differential role of species canopy types and functional traits in capturing these effects in large vs. small trees.

Key words: Bayesian model, functional traits, neighbourhood crowding, size class, soil nutrients, tree growth.

INTRODUCTION

Identifying the main ecological drivers of community assembly and dynamics is a central goal in ecology, notably to improve our fundamental understanding of species' distribution and coexistence (Chesson, 2000; HilleRisLambers *et al.*, 2012). Tree growth is an essential component of forest dynamics and ecosystem functioning (Bowman *et al.*, 2013; Duffy *et al.*, 2017). It can be shaped by both local abiotic and biotic conditions (Coates *et al.*, 2013; Fortunel *et al.*, 2018). Generally, trees grow faster with increasing soil fertility (Condit *et al.*, 2013; Lévesque *et al.*, 2016; Wright *et al.*, 2018; Zemunik *et al.*, 2018; Zhao and Zeng, 2019) and decreasing neighbourhood crowding (Uriarte *et al.*, 2004; Canham *et al.*, 2006; Rozendaal *et al.*, 2020). However, most previous studies have focused on a single life stage or pooled all life stages together (Zhu *et al.*,

2017; Umaña *et al.*, 2018), which may overlook the importance of phenotypic variation among individual trees due to ontogeny that can modify tree growth and forest dynamics (Miner *et al.*, 2005). Thus, we still have a poor understanding of how tree growth response to abiotic and biotic drivers vary between species-specific size classes in forest systems.

Tree growth response to soil nutrients may change with ontogeny, in part because requirements for limiting soil resources may differ between saplings and adult trees, as the cost of maintaining living tissues is greater in large individuals. Large trees tend to have a more developed root system and thereby better access to soil nutrients than small trees (Drake *et al.*, 2009). Large trees also have greater potential for nutrient storage (Boege and Marquis, 2005) and are thus potentially less sensitive to changes in soil nutrients than small trees (Li *et al.*, 2018).

However, few studies so far considered nutrient limitation among tree species across size classes (but see Baribault *et al.*, 2012).

The impacts of biotic interactions on tree growth may also vary across life stages, notably because competition for resources (e.g. light, soil nutrients) is size-asymmetric (Potvin and Dutilleul, 2009; DeMalach *et al.*, 2016). Large trees take up a disproportionate amount of resources relative to their size (Schwinning and Weiner, 1998), which can lead to large trees growing faster than small trees (Comita *et al.*, 2010; Niinemets, 2010). Previous work in tropical forests has shown that small trees experience greater reduction in tree growth via neighbourhood crowding effects than large trees (Fortunel *et al.*, 2018). Unlike tropical forests with high species diversity, temperate forests have lower species diversity. Thus, it is unclear whether this pattern holds in temperate forests, and there is still limited evidence for the joint influence of neighbourhood crowding and soil fertility on tree growth to vary with size classes (Baribault and Kobe, 2011).

General differences in tree growth may vary with species canopy types. Canopy species tend to grow faster than mid-canopy and understorey species (Onoda *et al.*, 2014), in part because of higher light availability in upper canopy layers (Canham *et al.*, 1994; Ishii *et al.*, 2004). In addition, species canopy types may exhibit contrasting growth responses to soil fertility because canopy species with greater rooting depth and root area may be less limited in infertile soils than understorey species with shallow roots (Aber *et al.*, 1989). Furthermore, canopy species may be more sensitive to crowding by neighbours since they tend to be more light-demanding, while understorey species can sustain growth in the shaded understorey (Onoda *et al.*, 2014). Yet, little is known about whether size class differences in tree growth response to both soil and neighbours vary between canopy types.

Functional traits can capture species differences in tree growth and its response to abiotic and biotic factors (Violle *et al.*, 2007). For instance, fast-growing species tend to exhibit an acquisitive strategy with high maximum height (Hmax), specific leaf area (SLA) and specific root length (SRL), but low wood density (WD) and seed mass (SM), while slow-growing species overall show a conservative strategy with the opposite trait pattern (Poorter *et al.*, 2008; Kramer-Walter *et al.*, 2016; Kunstler *et al.*, 2016). Acquisitive and fast-growing species tend to be favoured in fertile soils, while conservative and slow-growing species generally occur in infertile soils (Ordoñez *et al.*, 2009; Katabuchi *et al.*, 2012; Fortunel *et al.*, 2014). Furthermore, the growth of species with greater WD but lower SLA is less impacted by neighbourhood crowding, suggesting that conservative species are more resistant to neighbourhood interactions than acquisitive species (Fortunel *et al.*, 2016; Kunstler *et al.*, 2016). Plant traits can have contrasting effects on tree growth between tree size classes (Lusk and Warton, 2007; Fortunel *et al.*, 2020), but few studies so far included the size-component into tree growth. An examination of differences between size classes in a subtropical forest in Puerto Rico revealed that larger trees of species with low Hmax and SM were more sensitive to crowding than smaller trees (Lasky *et al.*, 2015). The generality of the role of traits in mediating tree growth response to soil and neighbours across size classes is awaiting further tests in different forest biomes.

Here, we combined 38 902 growth data over 15 years with species trait information in a 25-ha temperate forest in northeast China to examine differences between size classes in tree growth

response to soil fertility and neighbourhood crowding, and the relationships with species canopy types and functional traits. Specifically, we asked the following questions: (1) Do effects of soil fertility and neighbourhood crowding on tree growth vary between size classes? We predict that the growth of smaller trees would show a greater increase with soil fertility and steeper decline with neighbourhood crowding than that of larger trees. (2) Are the effects of soil fertility and neighbourhood crowding on tree growth consistent between canopy, mid-canopy and understorey species? We predict that understorey species would be more sensitive to soil nutrients, while mid-canopy and canopy species would be more sensitive to neighbourhood crowding. (3) Do traits mediate tree growth response to soil fertility and neighbourhood crowding between size classes? We predict that tree growth response to soil fertility would be exacerbated in species with greater Hmax, SLA and SRL but lower WD and SM, and that these responses would be stronger in smaller trees than larger trees. We also predict that tree growth reduction by neighbourhood crowding would be lessened in species with greater WD but lower SLA, and that these effects would be stronger in smaller trees than in larger trees.

MATERIALS AND METHODS

Study site

Our study was conducted in the old-growth temperate forest 25-ha (500 × 500 m) Changbaishan Forest Dynamics Plot (CBS FDP), located in the Changbaishan Nature Reserve, Jilin, China (42°12'N, 128°32'E). Mean annual rainfall is ~700 mm and most of this occurs from June to September (480–500 mm). Mean annual temperature is 2.8 °C, with an average minimum in January of −13.7 °C, and the average maximal in July is 19.63 °C (Yang *et al.*, 1985). The CBS FDP is relatively flat with elevation ranging from 791.8 to 809.5 m above sea level. Soils are classified as dark brown forest soil (Mollisols according to the American Soil Taxonomy Series, 1999) (Fig. 1).

Tree growth and size classes

The CBS FDP was set up in 2004. All free-standing woody stems with diameter at breast height 1.3 m from the ground (DBH) ≥ 1 cm were tagged, identified to species, mapped and measured. A total of 38 902 individuals were recorded, belonging to 52 species, 32 genera and 18 families (Hao *et al.*, 2008).

Tree censuses have been conducted every 5 years (in 2004, 2009, 2014 and 2019). We focused on trees alive between two consecutive censuses (i.e. 2004–2009, 2009–2014 and 2014–2019). We calculated annualized absolute diameter of growth, $g_{i,t+1}$ (mm year^{−1}), for each focal tree i between the two consecutive censuses t and $t + 1$ as:

$$g_{i,t+1} = \frac{DBH_{i,t+1} - DBH_{i,t}}{(Date_{t+1} - Date_t) / 365.25}$$

where DBH is the focal tree size, and Date is the census date.

For each focal tree at a given census interval, we identified its neighbouring trees within a radius of 10 m at the beginning of each census interval. Wang *et al.* (2010) found that tree species

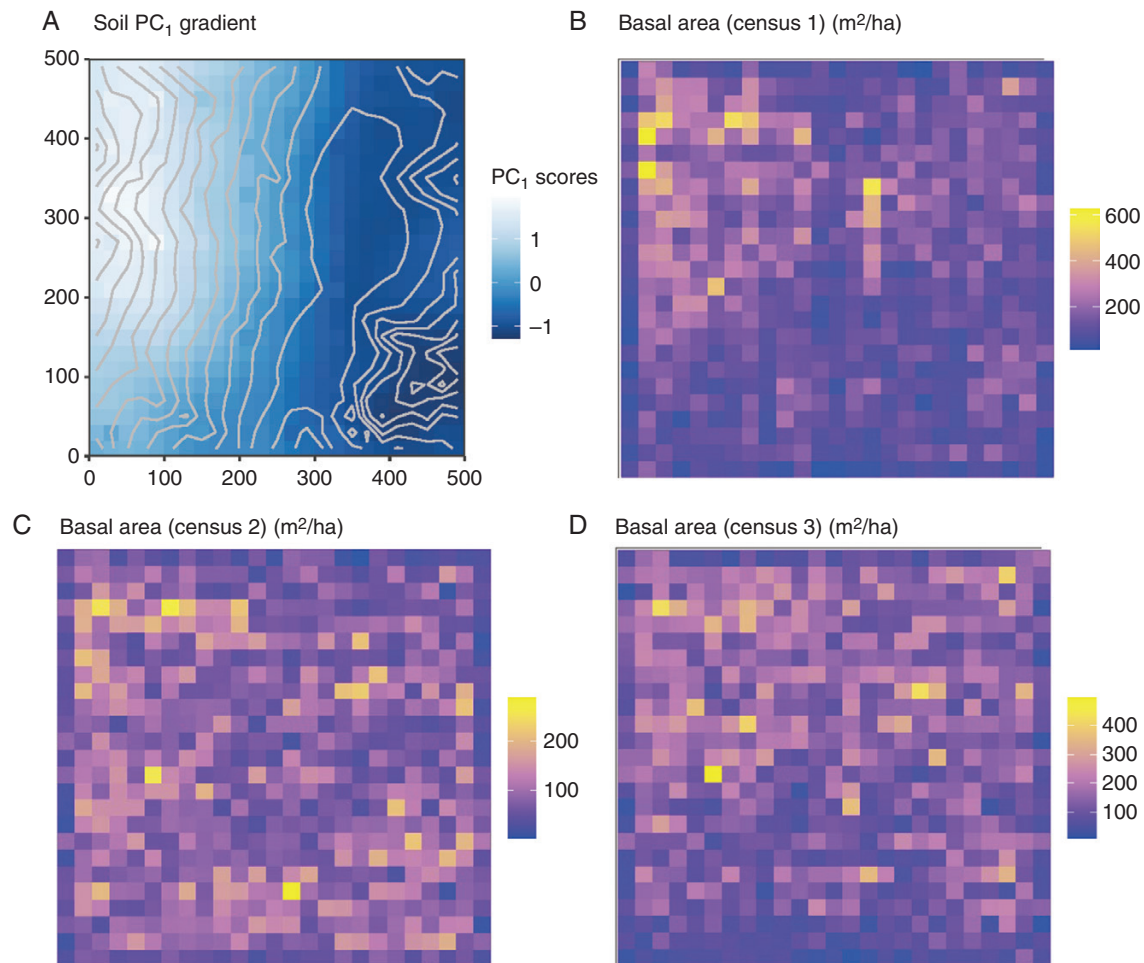


FIG. 1. Soil and forest structure gradients at the 20-m scale at the CBS FDP. (A) Soil gradient (using quadrat scores from soil PC₁, see [Supplementary Data Appendix S3](#)), and (B–D) basal area during the three studied census periods.

interactions disappeared beyond 20 m in our plot, so we chose a radius of 10 m for comparison with similar studies ([Lasky et al., 2014](#); [Fortunel et al., 2018](#)). To avoid edge effects, we excluded individuals <10 m from the plot boundaries from the focal trees. We also excluded: (1) broken main stems, (2) growth rates exceeding five standard deviations above the mean value and (3) individuals from four liana species found in our plot.

For each species at each census interval, we used the median DBH value to divide the individuals evenly into two size classes ([Lasky et al., 2015](#)): the small size class contains individuals with DBH lower than the median, and the large size class encompasses individuals with DBH greater than the median. For subsequent analysis we kept only focal species with more than three individuals in each size class (see [Supplementary Data Table S1](#) for details on the excluded species). The resulting dataset encompasses 78 957 growth records for 33 902 individuals belonging to 42 species ([Table 1](#)). We classified these 42 focal species into canopy, mid-canopy and understorey species ([Table 1](#)) based on available literature ([Wang et al., 2010](#); [Zhuang et al., 2014](#)) and expert knowledge at the study site ([Yang and Wu, 1988](#)).

Soil variables

In October 2007, soil samples were collected every 30 m, with two additional samples in random directions around these points at 2, 5 or 15 m ([Supplementary Data Figure S1](#)). All 967 soil samples were air-dried at ambient laboratory temperature before removing roots and stones using a 2-mm mesh sieve. Following [Lu \(1999\)](#), we measured soil pH, water content (SWC), organic matter (SOM), total nitrogen (Ntot), available nitrogen (Nav), total phosphorus (Ptot), available phosphorus (Pav), total potassium (Ktot) and available potassium (Kav). Soil pH in water (1 : 1) was determined using a Beckman glass electrode. SOM was measured with the dichromate oxidation colorimetric method. Ntot was estimated from the KCl extracts following the Kjeldahl method. Nav was determined with the alkali diffusion method using 1 mol L⁻¹ NaOH. Ptot was measured using H₂SO₄–HClO₄ following the molybdenum–antimony anti-spectrophotometric method. Pav was determined using a 0.05 mol L⁻¹ HCl – 0.025 mol L⁻¹ H₂SO₄. Ktot was determined by digestion in hydrofluoric acid and then using an atomic absorption spectrometer. Kav was extracted with 1 mol L⁻¹ NH₄Ac and detected by an atomic absorption spectrometer. Ktot and Kav were analysed with an atomic absorption

TABLE 1. List of focal 42 species, ranked by number of individuals and by canopy type (can: canopy, mid: mid-canopy, und: understory). Mean and range of DBH (cm) and annual growth rate (mm year⁻¹) are indicated by tree size class (small/large).

Species	Family	Abbreviation	Canopy type	Number of individuals	DBH			Growth		
					Small size class			Large size class		
<i>Tilia amurensis</i>	Tiliaceae	TILAMU	can	2313	20.03 [1.37, 4.00]	48.971 [37.50, 109.40]	1.363 [0.010, 7.940]	1.860 [0.010, 7.940]		
<i>Pinus koraiensis</i>	Pinaceae	PINKOR	can	2183	22.01 [1.80, 30.90]	43.78 [31.00, 97.00]	1.283 [0.010, 7.737]	2.216 [0.010, 7.126]		
<i>Ulmus japonica</i>	Ulmaceae	ULMJAP	can	917	3.06 [1.00, 6.70]	27.32 [6.80, 84.20]	1.411 [0.020, 6.311]	1.978 [0.060, 6.719]		
<i>Quercus mongolica</i>	Fagaceae	QUEMON	can	836	24.17 [1.00, 45.80]	60.58 [46.00, 108.00]	1.923 [0.010, 7.940]	2.469 [0.010, 7.940]		
<i>Fraxinus mandshurica</i>	Oleaceae	FRAMAN	can	600	32.78 [1.00, 52.50]	64.04 [52.80, 100.30]	2.688 [0.200, 7.940]	3.255 [0.200, 7.533]		
<i>Tilia mandshurica</i>	Tiliaceae	TILMAN	can	299	3.07 [1.00, 6.80]	21.42 [7.00, 76.50]	0.867 [0.010, 3.540]	1.383 [0.010, 5.604]		
<i>Ulmus laciniata</i>	Ulmaceae	ULMLAC	can	140	1.94 [1.00, 3.50]	11.12 [3.80, 53.60]	1.092 [0.036, 2.602]	2.144 [0.010, 6.805]		
<i>Populus davidiana</i>	Salicaceae	POPDV	can	24	14.66 [1.80, 21.00]	32.89 [21.10, 97.50]	1.926 [0.600, 6.002]	1.706 [0.200, 6.004]		
<i>Populus ussuriensis</i>	Salicaceae	POPUSS	can	22	85.03 [2.70, 113]	135.91 [119.00, 159.00]	4.072 [1.001, 7.805]	3.065 [1.001, 5.293]		
<i>Populus koreana</i>	Salicaceae	POPKOR	can	15	63.83 [38.50, 92.00]	119.97 [99.00, 139.00]	4.258 [0.407, 7.737]	2.097 [0.801, 3.402]		
<i>Abies holophylla</i>	Pinaceae	ABIHOL	can	6	27.07 [17.10, 41.60]	56.23 [45.20, 67.00]	3.289 [1.601, 4.803]	3.936 [3.002, 4.603]		
<i>Acer mono</i>	Aceraceae	ACEMON	mid	5679	1.89 [1.00, 3.24]	13.13 [3.29, 49.70]	0.885 [0.004, 4.418]	1.201 [0.002, 5.524]		
<i>Acer pseudo-sieboldianum</i>	Aceraceae	ACEPSE	mid	4274	3.05 [1.50, 4.00]	10.01 [5.45, 47.10]	1.013 [0.004, 5.230]	1.151 [0.002, 5.204]		
<i>Maackia amurensis</i>	Leguminosae	MAAAMU	mid	575	6.62 [1.00, 10.50]	15.33 [10.60, 35.30]	1.492 [0.010, 6.108]	1.798 [0.010, 6.719]		
<i>Acer tegmentosum</i>	Aceraceae	ACETEG	mid	573	1.68 [1.00, 2.80]	8.11 [2.90, 22.20]	1.609 [0.040, 6.448]	1.892 [0.010, 6.004]		
<i>Padus racemosa</i>	Rosaceae	PRUPAD	mid	388	1.72 [1.00, 2.70]	8.30 [2.80, 24.20]	1.574 [0.061, 5.650]	1.995 [0.002, 7.737]		
<i>Acer triflorum</i>	Aceraceae	ACETRI	mid	235	2.81 [1.00, 6.50]	15.57 [6.60, 37.40]	1.188 [0.010, 5.165]	1.162 [0.010, 4.403]		
<i>Acer mandshuricum</i>	Aceraceae	ACEMAN	mid	191	1.83 [1.00, 2.90]	12.66 [3.00, 32.70]	0.883 [0.060, 3.074]	1.101 [0.193, 3.302]		
<i>Malus baccata</i>	Rosaceae	MALBAC	mid	107	2.81 [1.60, 4.00]	19.45 [6.50, 48.30]	1.074 [0.032, 2.342]	1.463 [0.200, 5.904]		
<i>Betula platyphylla</i>	Betulaceae	BETPLA	mid	86	14.87 [8.70, 20.50]	28.45 [20.90, 46.40]	1.557 [0.204, 7.737]	3.261 [0.204, 7.126]		
<i>Rhamnus ussuriensis</i>	Rhamnaceae	RHAUSS	mid	83	4.25 [1.10, 6.60]	9.88 [6.70, 24.60]	1.648 [0.061, 7.533]	1.460 [0.020, 6.108]		
<i>Pyrus ussuriensis</i>	Rosaceae	PYRUSS	mid	54	2.73 [1.10, 6.00]	14.57 [6.10, 45.90]	1.156 [0.102, 3.339]	1.485 [0.200, 4.276]		
<i>Phellodendron amurense</i>	Rutaceae	PHEAMU	mid	51	15.94 [8.00, 20.70]	24.54 [21.10, 34.10]	2.391 [0.801, 6.719]	2.986 [0.407, 6.805]		
<i>Cerasus maximowiczii</i>	Rosaceae	CERMAX	mid	42	3.51 [1.20, 6.10]	10.40 [6.50, 19.00]	1.546 [0.468, 3.871]	1.281 [0.200, 4.072]		
<i>Sorbus alnifolia</i>	Rosaceae	SORALN	mid	23	3.06 [1.80, 5.20]	10.08 [5.90, 16.30]	1.026 [0.366, 1.893]	1.314 [0.407, 5.090]		
<i>Rhamnus davurica</i>	Rhamnaceae	RHADAV	mid	14	2.98 [1.20, 6.10]	10.28 [7.30, 13.80]	1.152 [0.180, 2.769]	1.353 [0.400, 2.443]		
<i>Betula costata</i>	Betulaceae	BETCOS	mid	12	11.76 [1.60, 19.90]	36.34 [25.20, 59.50]	1.476 [0.400, 5.204]	3.542 [0.200, 5.003]		
<i>Corylus mandshurica</i>	Betulaceae	CORMAN	und	5678	1.31 [1.00, 1.60]	2.11 [1.60, 4.60]	0.747 [0.002, 2.932]	0.619 [0.002, 3.040]		
<i>Acer barbinerve</i>	Aceraceae	ACEBAR	und	3105	1.51 [1.00, 2.00]	2.974 [2.01, 7.50]	0.759 [0.002, 3.909]	0.856 [0.002, 3.727]		
<i>Syringa reticulata</i>	Oleaceae	SYRRET	und	1188	2.28 [1.00, 3.50]	5.25 [3.60, 10.80]	1.187 [0.016, 5.293]	1.150 [0.020, 5.175]		
<i>Philadelphus schrenkii</i>	Saxifragaceae	PHISCH	und	245	1.09 [1.00, 1.20]	1.47 [1.30, 2.40]	0.527 [0.002, 2.402]	0.454 [0.020, 2.199]		
<i>Acer ginnala</i>	Aceraceae	ACEGIN	und	98	2.09 [1.00, 3.20]	5.51 [3.40, 9.10]	1.128 [0.020, 3.854]	1.326 [0.061, 3.380]		

TABLE 1. Continued

Species	Family	Abbreviation	Canopy type	Number of individuals	DBH		Growth	
					Small size class	Large size class	Small size class	Large size class
<i>Crataegus maximowiczii</i>	Rosaceae	CRAMAX	und	91	1.40 [1.00, 1.90]	2.88 [2.00, 4.90]	0.945 [0.024, 3.258]	0.840 [0.012, 2.508]
<i>Viburnum sargentii</i>	Caprifoliaceae	VIBSAR	und	32	1.2 [1.00, 1.40]	1.64 [1.50, 2.00]	0.832 [0.060, 1.881]	0.858 [0.102, 2.105]
<i>Euonymus pauciflorus</i>	Celastraceae	EUOPAU	und	30	1.14 [1.00, 1.30]	1.81 [1.40, 3.90]	0.772 [0.120, 1.501]	0.435 [0.081, 0.929]
<i>Euonymus alatus</i>	Celastraceae	EUOALA	und	22	1.16 [1.00, 1.40]	1.78 [1.50, 2.50]	0.891 [0.120, 1.663]	0.676 [0.280, 1.077]
<i>Lonicera chrysantha</i>	Caprifoliaceae	LONCHR	und	20	1.18 [1.10, 1.295]	1.67 [1.30, 2.10]	0.813 [0.181, 1.696]	0.473 [0.010, 1.161]
<i>Acanthopanax senticosus</i>	Araliaceae	ACASEN	und	16	1.04 [1.00, 1.10]	1.49 [1.20, 1.90]	0.708 [0.204, 1.425]	0.684 [0.224, 1.721]
<i>Viburnum burejaeticum</i>	Caprifoliaceae	VIBBUR	und	10	1.06 [1.00, 1.10]	1.42 [1.20, 2.00]	0.793 [0.047, 1.637]	0.584 [0.220, 0.969]
<i>Sambucus williamsii</i>	Caprifoliaceae	SAMWIL	und	6	1.68 [1.00, 2.20]	2.60 [2.20, 3.00]	1.585 [0.713, 3.950]	1.547 [0.672, 2.422]
<i>Aralia elata</i>	Araliaceae	ARAEAL	und	6	1.49 [1.20, 1.70]	3.00 [2.20, 4.10]	1.356 [0.040, 3.929]	2.727 [0.148, 5.961]
<i>Euonymus macropterus</i>	Celastraceae	EUOMAC	und	6	1.17 [1.00, 1.30]	1.65 [1.60, 1.70]	1.101 [0.814, 1.527]	0.214 [0.020, 0.407]

spectrometer (ELx808 Absorbance Microplate Reader, BioTek Instruments, Winooski, VT, USA). Further description can be found in Wang *et al.* (2012).

We used kriging methods to interpolate a soil map for each 10×10 -m quadrat. We used a principal component analysis (PCA) to evaluate covariations between the nine soil variables (Supplementary Data Appendices S3 and S4). The first two PC axes respectively explained 57.7 % and 17.8 % of the variation in soil data. Specifically, the PC₁ axis was positively correlated with Ntot, Nav and Ptot, and negatively with Ktot and Kav, whereas the PC₂ axis was positively associated with Pav, Ktot and Kav (Supplementary Data Figure S2). To test the effect of soil nutrients on tree growth hereafter, we used the following eight soil variables: quadrat scores on soil PC₁ and PC₂ axes, and quadrat values of soil Ntot, Nav, Ptot, Pav, Ktot and Kav that define the main axes of the soil PCA.

Functional traits

We used standard protocols (Cornelissen *et al.*, 2003) to measure five traits for all species in the CBS plot between July and September 2010 (Fang *et al.*, 2019): maximum height (Hmax, m), stem wood density (WD, g cm⁻³), specific leaf area (SLA, m² g⁻¹), specific root length (SRL, m g⁻¹) and seed mass (SM, g). These traits represent major functional dimensions of species ecological strategies and are closely associated with above- and below-ground resource acquisition and defence against natural enemies (Westoby, 1998; Wright *et al.*, 2004; Chave *et al.*, 2009; Weemstra *et al.*, 2016). Specifically, Hmax reflects tree stature and light capture ability (Moles *et al.*, 2009). For species with fewer than 20 individuals, we estimated Hmax as the tallest individual of these species in the CBS plot (2004 census). For species with over 20 individuals, we estimated Hmax as the mean of the 6–10 tallest individuals with greater DBH in 2004, and found that these estimates of Hmax were consistent with DBH_{max} calculated as the 95th quantile of the diameter distribution in three census periods. Stem WD relates to mechanical strength and defence against pests and pathogens (Chave *et al.*, 2009). As increment borers are prohibited in Changbai Nature Reserve, we collected 10 tree cores for each species in a nearby (10 km) Guangming Forest Farm. We measured fresh volume using the water displacement method, and weighed the dry mass after drying in the oven at 100 °C for 48 h. We calculated WD as the oven-dried mass divided by fresh volume (Wright *et al.*, 2010). SLA relates to light capture and leaf construction cost (Wright *et al.*, 2004). We collected and selected 12 fully expanded and intact canopy leaves (or 200–300 perennial needles for conifers) from the 6–12 largest individuals of each species in the CBS plot, and we then scanned each leaf with a portable scanner (Canon LiDE 110, Tokyo, Japan) before oven-drying at 60 °C for 48 h to get leaf dry mass. We calculated SLA as leaf area divided by leaf dry mass. SRL represents root ability to access soil nutrients (Comas and Eissenstat, 2004). We selected one adult individual per species, dug out the sub-roots that stemmed along the main root, and retrieved the root sample together with the surrounding soil with a shovel. Then we cleaned the soil particles on the roots with running water, picked first-order roots under a dissecting microscope and coloured their length before scanning using a portable scanner (Canon LiDE 110).

to measure root length with WinFOLIA (Régent Instruments, QC, Canada). We oven-dried root samples at 80 °C for at least 48 h before weighing. We calculated SRL as root length divided by root dry mass (Cornelissen *et al.*, 2003). SM relates to establishment success, competitive ability and tolerance to pests (Westoby *et al.*, 2002; Westoby and Wright, 2006). We sampled 30–200 mature and alive seeds in more than five individuals for each species, removed any accessory parts, then oven-dried them at 80 °C for at least 48 h before weighing the dry mass (Liu *et al.*, 2012). Detailed methods can be found in Fang *et al.* (2019).

Statistical analyses

We built a hierarchical Bayesian model of tree growth to test the effects of soil, neighbours and their interaction on tree growth.

To evaluate the effects of neighbours, we calculated a neighbourhood crowding index, $NCI_{t,s,i}$, for each focal stem i of species s based on the size (DBH) and distance (d) of its J_k neighbours of species K within a 10-m radius at the start of census interval t (Lasky *et al.*, 2014; Fortunel *et al.*, 2018):

$$NCI_{t,s,i} = \sum_{k=1}^K \sum_{\substack{j=1 \\ j \neq i}}^{J_k} \frac{DBH_{t,k,j}^2}{d_{k,j}}$$

We modelled expected tree growth for each stem i of species s at census interval t as:

$$\begin{aligned} E[\log(g_{t,s,i})] = & \beta_{1s} + \beta_{2s} \log(DBH_{t,s,i}) \\ & + \beta_{3s} \text{soil}_{s,i} + \beta_{4s} \log(NCI_{t,s,i}) \\ & + \beta_{5s} \text{soil}_{s,i} \log(NCI_{t,s,i}) + \beta_{6s} \log \\ & (DBH_{t,s,i}) \log(NCI_{t,s,i}) + \delta_i + \gamma_i + \sigma_i \end{aligned}$$

where $g_{t,s,i}$ represents the annualized absolute diameter growth rate, $DBH_{t,s,i}$ is the stem diameter at 1.3 m height at the start of the census interval, $\text{Soil}_{s,i}$ is the soil variable of the quadrat where the focal stem is found and $NCI_{t,s,i}$ is the neighbourhood crowding of the focal stem at the start census interval. Fixed effects for year and random effects for individual tree and quadrat are denoted as δ_i , γ_i and σ_i respectively.

To test how traits affect (1) tree growth and its response to (2) soil variables, (3) neighbourhood crowding, (4) the interaction between soil nutrients and neighbourhood crowding, and (5) the interaction between tree size and neighbourhood crowding, we used five second-level regressions:

$$\beta_{cs} = \beta_{c,0} + \beta_{c,1} F_s + \varepsilon_{cs}$$

where $\beta_{c,0}$ represents the mean value across all species for the model covariate c , and $\beta_{c,1}$ represents departure from the mean effect with one standard deviation change in the value of trait F_s of species s , and ε_{cs} is the normally distributed random effect of species s .

We standardized all model covariates to enable direct comparisons of the relative importance of each effect. We standardized soil variables by subtracting the mean and dividing by one standard deviation computed across the dataset. DBH and NCI were log-transformed prior to standardization. We standardized

DBH and NCI by subtracting species-specific means and dividing by species-specific standard deviations. All traits except WD were log-transformed prior to scaling, and we scaled all traits with mean = 0 and standard deviation = 1.

For each of the two tree size classes (small vs. large), we fit separate models for (1) each of the eight soil variables (soil PC_1 and PC_2 , N_{tot} , N_{av} , P_{tot} , P_{av} , K_{tot} and K_{av}) and (2) each of the five traits (H_{max} , WD, SLA, SRL and SM), giving a total of 80 models. We set diffuse priors and estimated posterior sampling via Markov chain Monte Carlo routines conducted in JAGS 3.4.0. Most chains converged with 80 000 iterations (Supplementary Data Table S3). We computed the median and the 2.5 and 97.5 percentiles to obtain 95 % credible intervals of each model parameter (Supplementary Data Table S3), and evaluated the goodness of fit of each model (Supplementary Data Table S4). We extracted species-specific beta parameters and used two-way ANOVA to test for differences in soil and neighbour effects on tree growth between canopy, mid-storey and understorey species.

All analyses were conducted in the R 3.6.2 statistical platform using packages psych (Revelle, 2011), r2jags (Su and Yajima, 2015) and smatr (Warton *et al.*, 2012).

RESULTS

Differences in tree growth response to soil and crowding between tree size classes

Among all eight studied soil variables, only soil PC_1 (i.e. increasing total N and P) significantly influenced tree growth, so we focus hereafter on results from this model (results for the other seven soil variables are available in Supplementary Data Table S3). Small trees grew faster with increasing soil total N and P, but large trees showed no significant response to soil gradients ($\beta_{3,0}$ parameter; Fig. 2; Supplementary Data Figure S3). Small trees of understorey species exhibited a stronger growth response to soil PC_1 than small trees of canopy and mid-storey species, while we detected no differences in large tree growth response to soil PC_1 between canopy types (Fig. 3B).

Neighbourhood crowding consistently reduced tree growth in both tree size classes, and this effect was stronger in large than small trees ($\beta_{4,0}$ parameter; Fig. 2; Supplementary Data Figure S3). For both small and large trees, the negative effect of neighbourhood crowding on tree growth was strongest in canopy species and lowest in understorey species, with mid-canopy species exhibiting an intermediate growth response (Fig. 3C).

In both tree size classes, tree growth generally showed no response to the soil–neighbourhood and size–neighbourhood interactions ($\beta_{5,0}$ and $\beta_{6,0}$ parameters; Fig. 2; Supplementary Data Figure S3). However, the soil–neighbourhood interaction more strongly reduced tree growth in small trees of canopy species than of understorey and mid-canopy species, while differences in tree growth of large trees between canopy types were limited (Fig. 3D).

Finally, tree growth increased with tree size in the small tree size class, but not in the large tree size class ($\beta_{2,0}$ parameter, Fig. 2; Supplementary Data Figure S3). Large trees of canopy species grew faster with increasing tree size compared

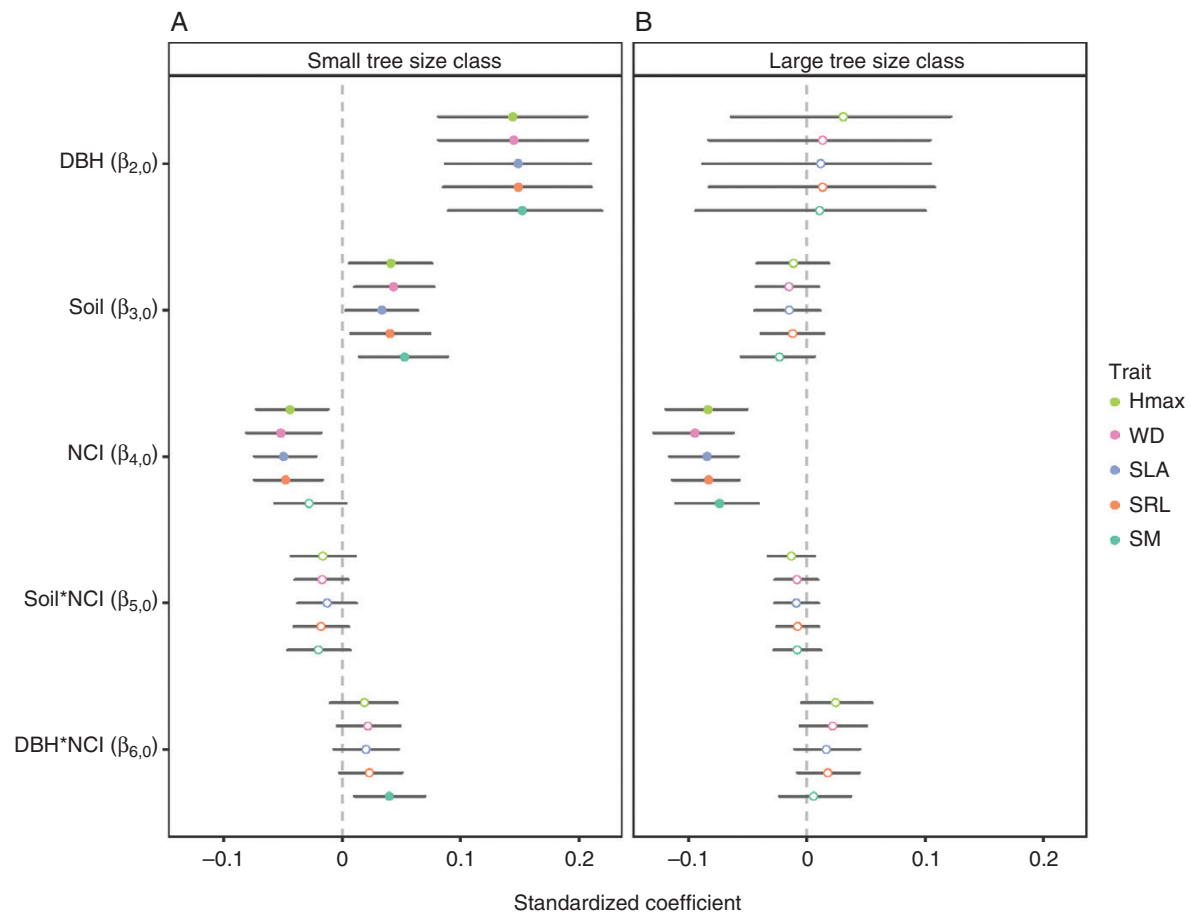


FIG. 2. Effects of tree size (DBH, parameter $\beta_{2,0}$), soil (Soil, parameter $\beta_{3,0}$), neighbourhood crowding (NCI, parameter $\beta_{4,0}$), the soil–neighbourhood interaction (Soil*NCI, parameter $\beta_{5,0}$), and the tree size–neighbourhood interaction (DBH*NCI, parameter $\beta_{6,0}$) on tree growth of (A) small and (B) large tree size classes. Abbreviations: Hmax, maximum size; WD, wood density; SLA, specific leaf area; SRL, specific root length; SM, seed mass.

to those of mid-canopy and understorey species (Fig. 3A). In the large tree size class, relatively larger trees were less impacted by neighbours in canopy species than in mid-canopy species (Fig. 3E).

Trait effects on growth response to soil and crowding between tree size classes

Traits effects on tree growth were generally consistent across the two tree size classes ($\beta_{1,1}$ parameter; Fig. 4; Supplementary Data Figure S4). Specifically, species with high Hmax but low WD and SLA exhibited faster growth, and these trait effects were stronger in large than small trees ($\beta_{1,1}$ parameter; Fig. 4; Supplementary Data Figure S4).

Overall, traits did not significantly influence tree growth response to soil nutrients ($\beta_{3,1}$ parameter; Fig. 4; Supplementary Data Figure S4), though smaller trees with higher SLA had a stronger growth response to soil PC_1 ($\beta_{3,1}$ parameter; Fig. 4; Supplementary Data Figure S4).

Conversely, some traits mediated tree growth response to neighbourhood crowding ($\beta_{4,1}$ parameter; Fig. 4; Supplementary Data Figure S4). Specifically, for both tree size classes, the growth of species with higher SM was further reduced by

neighbourhood crowding. For large trees, the growth of species with high SRL and SLA was less affected by neighbourhood crowding.

Traits did not influence tree growth response to the soil–neighbourhood interaction ($\beta_{5,1}$ parameter; Fig. 4; Supplementary Data Figure S4). However, some traits mediated tree growth response to the DBH–crowding interaction ($\beta_{6,1}$ parameter; Fig. 4; Supplementary Data Figure S4). In particular, the growth of relatively larger trees from species with higher SM had a weaker response to the DBH–crowding interaction compared with the small tree size class, but was more sensitive to crowding compared with the small tree size class.

DISCUSSION

Elucidating the role of ecological drivers on tree growth across life stages represents a step towards a better understanding of forest dynamics (Adler *et al.*, 2013; Piao *et al.*, 2013). Here, combining census data over 15 years for 42 temperate tree species with spatially explicit soil and neighbourhood crowding information in a temperate forest in northeastern China, we uncover variation in the relative importance of abiotic and biotic factors on tree growth across size classes.

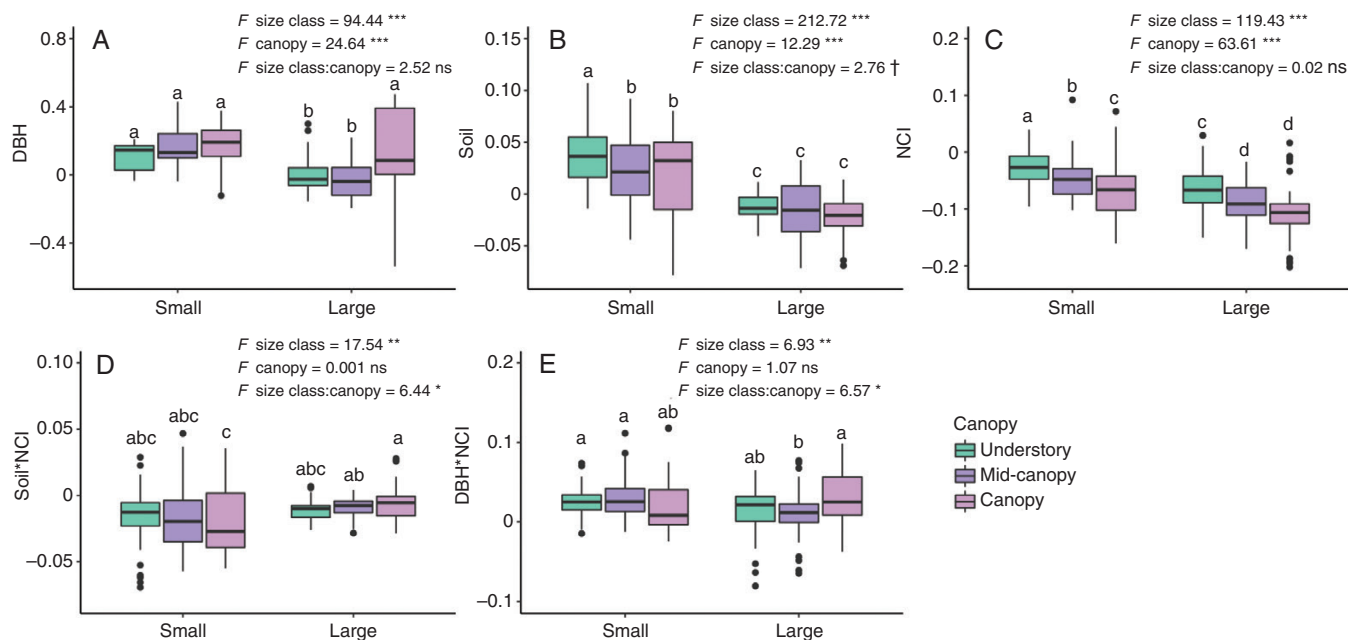


FIG. 3. Differences in species-specific effects of tree size (DBH, parameter $\beta_{2,0}$), soil (Soil, parameter $\beta_{3,0}$), neighbourhood crowding (NCI, parameter $\beta_{4,0}$), the soil–neighbourhood interaction (Soil*NCI, parameter $\beta_{5,0}$), and tree size–neighbourhood interaction (DBH*NCI, parameter $\beta_{6,0}$) between the two size classes and among three canopy types. Two-way ANOVA *F* statistics are shown with significance test (ns, non-significant; †, $0.05 < P < 0.1$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$); *post hoc* Tukey's HSD (honestly significant difference) test results are indicated using bold letters.

Size class change in growth response to soil nutrients and neighbourhood crowding

We show that tree growth increased with increasing soil total N and P in the small tree size class but not in the large tree size class, suggesting that large trees are less limited by soil nutrients than small trees. Our findings are in line with previous work that used fertilization experiments to examine tree growth response to nutrient addition. In subtropical, warm and cold temperate forests, small trees exhibited faster growth with P addition, while large trees showed no response (Li *et al.*, 2018). In a lowland tropical forest, saplings and poles ($1 \leq \text{DBH} < 10$ cm) grew faster with joint addition of K and N, while large trees exhibited a marginally significant response (Wright *et al.*, 2011). However, most previous studies used a single threshold – e.g. 15 cm in Li *et al.* (2018) and 10 cm in Wright *et al.* (2011) – to define the small and large size classes, without accounting for differences in tree size distributions between species or their canopy types. Our results show that the growth response of small trees of understorey species to soil nutrients is stronger than that of small trees of canopy and mid-canopy species, suggesting that soil fertility may play an important role in early life stages of understorey species. This effect is stronger in understorey species than in mid-canopy and canopy species, which indicates that understorey species may be poor competitors for soil resource acquisition.

In addition, we show that neighbourhood crowding strongly reduces tree growth across both tree size classes, in line with previous studies (Comita *et al.*, 2010; Fortunel *et al.*, 2018). Contrary to our expectation, we find that the neighbourhood crowding effect reduces tree growth more steeply in large than in small trees, though this result hides differences between canopy types. Interestingly, our study uncovers that, across tree

size classes, the growth of canopy species is more strongly reduced by neighbours than that of understorey species, in line with the idea that canopy species are more resource-limited than understorey species (Niinemets, 2010). Our findings thus highlight that size-asymmetric competition for limiting nutrients such as light can vary not only with tree size but also with species canopy types as trees move between canopy layers (Pacala *et al.*, 1996).

While we find that soil and neighbours separately impact tree growth at the CBS FDP, overall we detect no effect of their interaction. Our results suggest that environmental filtering and biotic interactions may act mainly independently in shaping tree growth in this temperate forest (Zhang *et al.*, 2016), as also found in a tropical forest (Fortunel *et al.*, 2018). However, this interaction more strongly reduces the growth of small canopy trees than of understorey and mid-canopy species, which suggests that young stages of canopy species may be particularly sensitive to increased crowding effects at higher soil fertility (Coomes *et al.*, 2009), as predicted from the stress gradient hypothesis (Bertness and Callaway, 1994).

Role of traits in tree growth response to soil and neighbours with ontogeny

Traits consistently influence tree growth across small and large trees. As expected, species with lower WD and higher Hmax grow faster, consistent with low volumetric construction costs (Wright *et al.*, 2010) and greater light capture (Davies *et al.*, 1998), respectively. However, species with lower SLA exhibit faster growth, which does not align with previous studies (Wright *et al.*, 2010; Lasky *et al.*, 2015). This may stem from the fact that the faster-growing species at our study site are

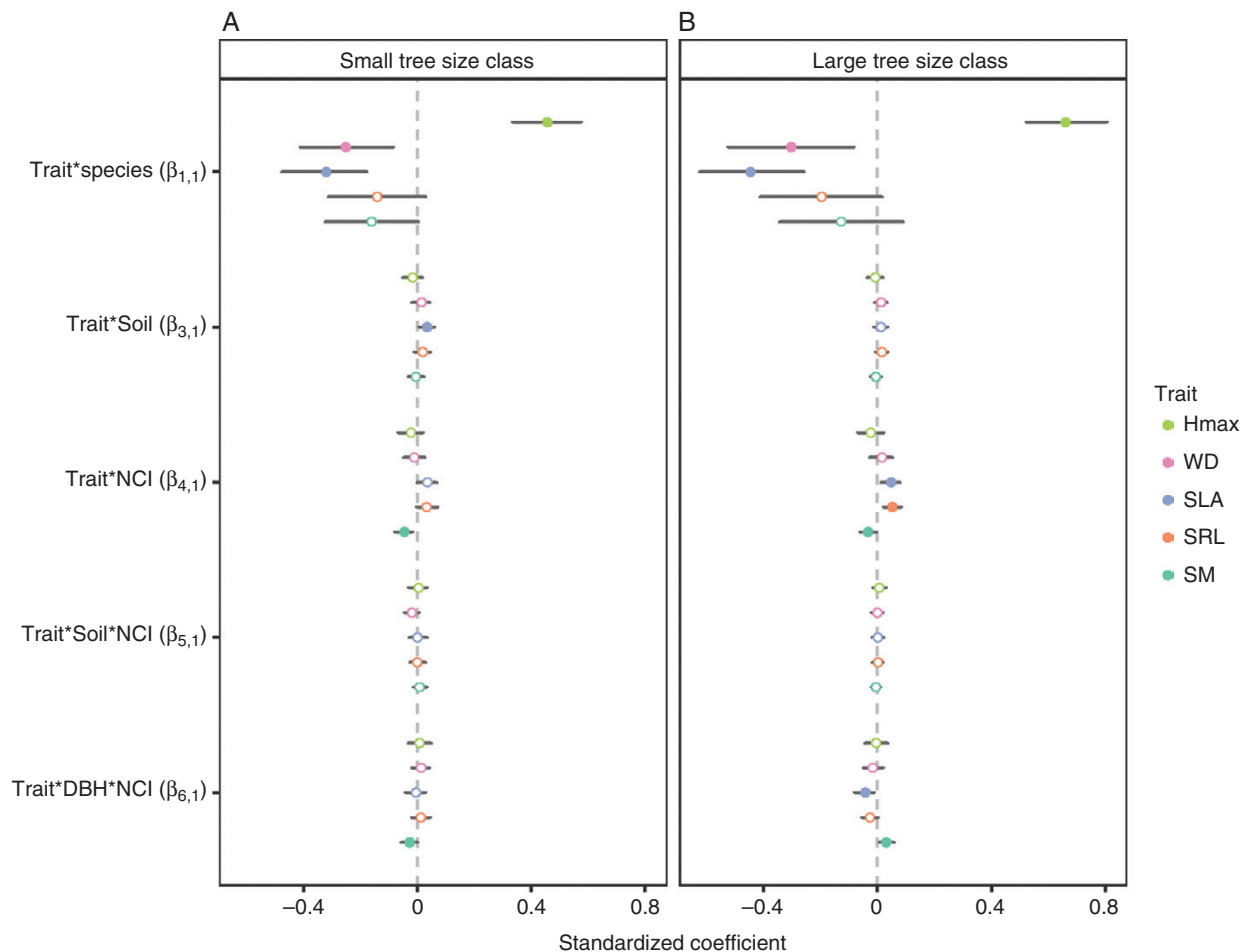


FIG. 4. Effects of functional traits on tree growth response to soil (Soil, parameter $\beta_{2,1}$), neighbourhood crowding (NCI, parameter $\beta_{3,1}$), the soil–neighbourhood interaction (Soil*NCI, parameter $\beta_{4,1}$) and the tree size–neighbourhood interaction (DBH*NCI, parameter $\beta_{5,1}$) in (A) small and (B) large tree size classes. Abbreviations: Hmax, maximum size; WD, wood density; SLA, specific leaf area; SRL, specific root length; SM, seed mass.

conifer species (together representing 7.2 % of the plot stems), with typically low SLA values.

We find limited evidence of trait-mediated soil effects on tree growth, though, as expected, we did detect the role of traits in small rather than large trees. Our results are consistent with previous work showing that larger trees have greater associations between growth and environment than saplings because environmental filtering has eliminated individuals with traits that are less well suited to local conditions (Webb and Peart, 2000; Cornwell *et al.*, 2006; Kraft *et al.*, 2008; Shen *et al.*, 2014). In particular, we demonstrate that small trees of species with higher SLA grow even faster with increasing soil fertility, suggesting that species that capture light more efficiently can also make better use of increased soil nutrient availability (Ordoñez *et al.*, 2009).

We find some support for trait-mediated neighbourhood crowding on tree growth. In particular, large trees of species with high SLA and SRL are less sensitive to crowding. Our findings are consistent with previous studies conducted in a tropical forest (Lasky *et al.*, 2015), where large individuals of species that invest in resource capture (e.g. light and soil nutrients) suffer less from competition by their neighbours. Conversely, we find that both small and large trees of species

with higher seed mass were more sensitive to neighbourhood crowding, suggesting that the tolerance–fecundity trade-off can affect tree growth throughout ontogenetic stages (Muller-Landau Helene, 2010; Ejsmond *et al.*, 2019).

Overall, we show that traits do not mediate tree growth response to the interaction between soil nutrients and crowding, in line with previous research in a tropical forest (Fortunel *et al.*, 2018). One possible explanation is that the relatively slow tree dynamics in our old-growth temperate forest compared to secondary tropical forests may make it more difficult to capture the role of traits (Lasky *et al.*, 2014). In addition, the role of traits may vary with the type of biotic interactions influencing tree growth response to soil fertility (Fortunel *et al.*, 2018).

Finally, we find some evidence of size class shifts in tree growth response to the interaction between tree size and crowding. Species with higher SM have a stronger growth response to the size–crowding interaction in the small tree size class, but have the opposite pattern in the large tree size class. This reversal may be consistent with the biomass allocation theory (Garnier, 1991; Poorter *et al.*, 2012), whereby during ontogeny larger plants allocate more resources to reproduction at the expense of growth, thus potentially reducing their ability

to mitigate neighbourhood crowding effects. We mostly sampled traits from adult, canopy trees at the CBS FDP, but traits can exhibit strong variation between developmental stages, in particular in leaves and wood (e.g. [Fortunel et al., 2020](#)). A stronger test of the role of traits on tree growth may require trait measurement at the corresponding life stage, associated with more in-depth characterization of tree ontogenetic trajectories from architectural characteristics, for each species ([Yang et al., 2018](#); [Asefa et al., 2021](#)).

SUPPLEMENTARY DATA

Supplementary data are available online at <https://academic.oup.com/aob> and consist of the following. **Table S1.** List of the discarded species from small and large size classes. **Figure S1.** Soil sampling at the Changbaishan Forest Dynamics Plot. **Figure S2.** Principal component analysis of soil variables. **Table S2.** Spearman correlation coefficients among soil variables. **Table S3.** Point-estimates, median values and 95 % credible intervals of tree growth model parameters. **Table S4.** Goodness of fit of tree growth models. **Figure S3.** Standardized regression coefficients for the community-level effects of tree size, soil, neighbours, soil–crowding interaction, and tree size–crowding interaction on tree growth in the small and large size classes. **Figure S4.** Standardized regression coefficients for the species-level effects of traits in mediating tree growth response to soil, neighbourhood crowding, soil–crowding interaction, and DBH–crowding interaction in small and large tree size classes.

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CONFLICT OF INTEREST

The authors declare there are no competing interests.

LITERATURE CITED

- Aber JD, Nadelhoffer KJ, Steudler P, Melillo JM. 1989. Nitrogen saturation in northern forest ecosystems: excess nitrogen from fossil fuel combustion may stress the biosphere. *BioScience* **39**: 378–386. doi:10.2307/1311067.
- Adler PB, Fajardo A, Kleinhesselink AR, Kraft NJB. 2013. Trait-based tests of coexistence mechanisms. *Ecology Letters* **16**: 1294–1306. doi:10.1111/ele.12157.
- Asefa M, Song X, Cao M, Lasky JR, Yang J. 2021. Temporal trait plasticity predicts the growth of tropical trees. *Journal of Vegetation Science* **32**: e13056.
- Baribault TW, Kobe RK. 2011. Neighbour interactions strengthen with increased soil resources in a northern hardwood forest. *Journal of Ecology* **99**: 1358–1372. doi:10.1111/j.1365-2745.2011.01862.x.
- Baribault TW, Kobe RK, Finley AO. 2012. Tropical tree growth is correlated with soil phosphorus, potassium, and calcium, though not for legumes. *Ecological Monographs* **82**: 189–203. doi:10.1890/11-1013.1.
- Bertness MD, Callaway R. 1994. Positive interactions in communities. *Trends in Ecology & Evolution* **9**: 191–193. doi:10.1016/0169-5347(94)90088-4.
- Boege K, Marquis RJ. 2005. Facing herbivory as you grow up: the ontogeny of resistance in plants. *Trends in Ecology & Evolution* **20**: 441–448.
- Bowman DMJS, Brien RJW, Gloor E, Phillips OL, Prior LD. 2013. Detecting trends in tree growth: not so simple. *Trends in Plant Science* **18**: 11–17. doi:10.1016/j.tplants.2012.08.005.
- Canham CD, Finzi AC, Pacala SW, Burbank DH. 1994. Causes and consequences of resource heterogeneity in forests: interspecific variation in light transmission by canopy trees. *Canadian Journal of Forest Research* **24**: 337–349. doi:10.1139/x94-046.
- Canham CD, Papaik MJ, Uriarte M, McWilliams WH, Jenkins JC, Twery MJ. 2006. Neighborhood analyses of canopy tree competition along environmental gradients in New England forests. *Ecological Applications* **16**: 540–554. doi:10.1890/1051-0761(2006)016[0540:naocte]2.0.co;2.
- Chave J, Coomes D, Jansen S, Lewis SL, Swenson NG, Zanne AE. 2009. Towards a worldwide wood economics spectrum. *Ecology Letters* **12**: 351–366. doi:10.1111/j.1461-0248.2009.01285.x.
- Chesson P. 2000. Mechanisms of maintenance of species diversity. *Annual Review of Ecology and Systematics* **31**: 343–366. doi:10.1146/annurev.ecolsys.31.1.343.
- Coates KD, Lilles EB, Astrup R. 2013. Competitive interactions across a soil fertility gradient in a multispecies forest. *Journal of Ecology* **101**: 806–818. doi:10.1111/1365-2745.12072.
- Comas LH, Eissenstat DM. 2004. Linking fine root traits to maximum potential growth rate among 11 mature temperate tree species. *Functional Ecology* **18**: 388–397. doi:10.1111/j.0269-8463.2004.00835.x.
- Comita LS, Muller-Landau HC, Aguilar S, Hubbell SP. 2010. Asymmetric density dependence shapes species abundances in a tropical tree community. *Science* **329**: 330–332. doi:10.1126/science.1190772.
- Condit R, Engelbrecht BMJ, Pino D, Pérez R, Turner BL. 2013. Species distributions in response to individual soil nutrients and seasonal drought across a community of tropical trees. *Proceedings of the National Academy of Sciences* **110**: 5064–5068. doi:10.1073/pnas.1218042110.
- Coomes DA, Kunstler G, Canham CD, Wright E. 2009. A greater range of shade-tolerance niches in nutrient-rich forests: an explanation for positive richness–productivity relationships? *Journal of Ecology* **97**: 705–717. doi:10.1111/j.1365-2745.2009.01507.x.
- Cornelissen JHC, Lavorel S, Garnier E, et al. 2003. A handbook of protocols for standardised and easy measurement of plant functional traits worldwide. *Australian Journal of Botany* **51**: 335–380. doi:10.1071/bt02124.
- Cornwell WK, Schwillk DW, Ackerly DD. 2006. A trait-based test for habitat filtering: convex hull volume. *Ecology* **87**: 1465–1471. doi:10.1890/0012-9658(2006)87[1465:atthfj]2.0.co;2.
- Davies SJ, Palmiotto PA, Ashton PS, Lee HS, Lafrankie JV. 1998. Comparative ecology of 11 sympatric species of *Macaranga* in Borneo: tree distribution in relation to horizontal and vertical resource heterogeneity. *Journal of Ecology* **86**: 662–673. doi:10.1046/j.1365-2745.1998.00299.x.
- DeMalach N, Zaady E, Weiner J, Kadmon R. 2016. Size asymmetry of resource competition and the structure of plant communities. *Journal of Ecology* **104**: 899–910. doi:10.1111/1365-2745.12557.
- Drake PL, Mendham DS, White DA, Ogden GN. 2009. A comparison of growth, photosynthetic capacity and water stress in *Eucalyptus globulus* coppice regrowth and seedlings during early development. *Tree Physiology* **29**: 663–674. doi:10.1093/treephys/tpp006.
- Duffy JE, Godwin CM, Cardinale BJ. 2017. Biodiversity effects in the wild are common and as strong as key drivers of productivity. *Nature* **549**: 261–264. doi:10.1038/nature23886.
- Ejsmond A, Kozłowski J, Ejsmond MJ. 2019. Probing of mortality rate by staying alive: The growth–reproduction trade-off in a spatially heterogeneous environment. *Functional Ecology* **33**: 2327–2337. doi:10.1111/1365-2435.13442.
- Fang S, Munoz F, Ye J, et al. 2019. Deterministic processes drive functional and phylogenetic temporal changes of woody species in temperate forests in Northeast China. *Annals of Forest Science* **76**: 42.
- Fortunel C, Ruelle J, Beauchêne J, Fine PVA, Baraloto C. 2014. Wood specific gravity and anatomy of branches and roots in 113 Amazonian

- rainforest tree species across environmental gradients. *New Phytologist* **202**: 79–94.
- Fortunel C, Valencia R, Wright SJ, Garwood NC, Kraft NJB. 2016. Functional trait differences influence neighbourhood interactions in a hyperdiverse Amazonian forest. *Ecology Letters* **19**: 1062–1070. doi:10.1111/ele.12642.
- Fortunel C, Lasky JR, Uriarte M, et al. 2018. Topography and neighborhood crowding can interact to shape species growth and distribution in a diverse Amazonian forest. *Ecology* **99**: 2272–2283. doi:10.1002/ecy.2441.
- Fortunel C, Stahl C, Heuret P, Nicolini E, Baraloto C. 2020. Disentangling the effects of environment and ontogeny on tree functional dimensions for congeneric species in tropical forests. *New Phytologist* **226**: 385–395. doi:10.1111/nph.16393.
- Garnier E. 1991. Resource capture, biomass allocation and growth in herbaceous plants. *Trends in Ecology & Evolution* **6**: 126–131. doi:10.1016/0169-5347(91)90091-B.
- Hao ZQ, Li BH, Zhang J, Wang XG, Ye J, Yao XL. 2008. Broad-leaved Korean pine (*Pinus koraiensis*) mixed forest plot in Changbaishan (CBS) of China: community composition and structure. *Journal of Plant Ecology* **32**: 238–250.
- HilleRisLambers J, Adler PB, Harpole WS, Levine JM, Mayfield MM. 2012. Rethinking community assembly through the lens of coexistence theory. *Annual Review of Ecology, Evolution, and Systematics* **43**: 227–248. doi:10.1146/annurev-ecolsys-110411-160411.
- Ishii HT, Tanabe S-i, Hiura T. 2004. Exploring the relationships among canopy structure, stand productivity, and biodiversity of temperate forest ecosystems. *Forest Science* **50**: 342–355.
- Katabuchi M, Kurokawa H, Davies SJ, Tan S, Nakashizuka T. 2012. Soil resource availability shapes community trait structure in a species-rich dipterocarp forest. *Journal of Ecology* **100**: 643–651.
- Kraft NJB, Valencia R, Ackerly DD. 2008. Functional traits and niche-based tree community assembly in an Amazonian forest. *Science* **322**: 580–582. doi:10.1126/science.1160662.
- Kramer-Walter KR, Bellingham PJ, Millar TR, Smissen RD, Richardson SJ, Laughlin DC. 2016. Root traits are multidimensional: specific root length is independent from root tissue density and the plant economic spectrum. *Journal of Ecology* **104**: 1299–1310. doi:10.1111/1365-2745.12562.
- Kunstler G, Falster D, Coomes DA, et al. 2016. Plant functional traits have globally consistent effects on competition. *Nature* **529**: 204–207. doi:10.1038/nature16476.
- Lasky JR, Uriarte M, Boukili VK, Chazdon RL. 2014. Trait-mediated assembly processes predict successional changes in community diversity of tropical forests. *Proceedings of the National Academy of Sciences* **111**: 5616–5621. doi:10.1073/pnas.1319342111.
- Lasky JR, Bachelot B, Muscarella R, et al. 2015. Ontogenetic shifts in trait-mediated mechanisms of plant community assembly. *Ecology* **96**: 2157–2169. doi:10.1890/14-1809.1.
- Lévesque M, Walthert L, Weber P. 2016. Soil nutrients influence growth response of temperate tree species to drought. *Journal of Ecology* **104**: 377–387. doi:10.1111/1365-2745.12519.
- Li Y, Tian D, Yang H, Niu S. 2018. Size-dependent nutrient limitation of tree growth from subtropical to cold temperate forests. *Functional Ecology* **32**: 95–105.
- Liu XJ, Swenson NG, Wright SJ, et al. 2012. Covariation in plant functional traits and soil fertility within two species-rich forests. *PLoS One* **7**: e34767. doi:10.1371/journal.pone.0034767.
- Lu RS. 1999. *Analytical methods of soil agrochemistry (in Chinese)*. Beijing: China Agricultural Science Technology Press.
- Lusk CH, Warton DI. 2007. Global meta-analysis shows that relationships of leaf mass per area with species shade tolerance depend on leaf habit and ontogeny. *New Phytologist* **176**: 764–774. doi:10.1111/j.1469-8137.2007.02264.x.
- Miner BG, Sultan SE, Morgan SG, Padilla DK, Relyea RA. 2005. Ecological consequences of phenotypic plasticity. *Trends in Ecology & Evolution* **20**: 685–692. doi:10.1016/j.tree.2005.08.002.
- Moles AT, Warton DI, Warman L, et al. 2009. Global patterns in plant height. *Journal of Ecology* **97**: 923–932.
- Muller-Landau Helene C. 2010. The tolerance–fecundity trade-off and the maintenance of diversity in seed size. *Proceedings of the National Academy of Sciences* **107**: 4242–4247.
- Niinemets U. 2010. Responses of forest trees to single and multiple environmental stresses from seedlings to mature plants: past stress history, stress interactions, tolerance and acclimation. *Forest Ecology and Management* **260**: 1623–1639. doi:10.1016/j.foreco.2010.07.054.
- Onoda Y, Saluñga JB, Akutsu K, Aiba S-i, Yahara T, Anten NPR. 2014. Trade-off between light interception efficiency and light use efficiency: implications for species coexistence in one-sided light competition. *Journal of Ecology* **102**: 167–175.
- Ordoñez JC, van Bodegom PM, Witte J-PM, Wright IJ, Reich PB, Aerts R. 2009. A global study of relationships between leaf traits, climate and soil measures of nutrient fertility. *Global Ecology and Biogeography* **18**: 137–149. doi:10.1111/j.1466-8238.2008.00441.x.
- Pacala SW, Canham CD, Saponara J, SilanderJAJr, Kobe RK, Ribbens E. 1996. Forest models defined by field measurements: estimation, error analysis and dynamics. *Ecological Monographs* **66**: 1–43. doi:10.2307/2963479.
- Piao T, Comita LS, Jin G, Kim JH. 2013. Density dependence across multiple life stages in a temperate old-growth forest of northeast China. *Oecologia* **172**: 207–217. doi:10.1007/s00442-012-2481-y.
- Poorter L, Wright SJ, Paz H, et al. 2008. Are functional traits good predictors of demographic rates? Evidence from five neotropical forests. *Ecology* **89**: 1908–1920. doi:10.1890/07-0207.1.
- Poorter H, Niklas KJ, Reich PB, Oleksyn J, Poot P, Mommer L. 2012. Biomass allocation to leaves, stems and roots: meta-analyses of interspecific variation and environmental control. *New Phytologist* **193**: 30–50.
- Potvin C, Dutilleul P. 2009. Neighborhood effects and size-asymmetric competition in a tree plantation varying in diversity. *Ecology* **90**: 321–327. doi:10.1890/08-0353.1.
- Revelle W. 2011. An overview of the psych package.
- Rozendaal DMA, Phillips OL, Lewis SL, et al. 2020. Competition influences tree growth, but not mortality, across environmental gradients in Amazonia and tropical Africa. *Ecology* **101**: e03052. doi:10.1002/ecy.3052.
- Schwinning S, Weiner J. 1998. Mechanisms determining the degree of size asymmetry in competition among plants. *Oecologia* **113**: 447–455. doi:10.1007/s004420050397.
- Shen Y, Santiago LS, Shen H, et al. 2014. Determinants of change in subtropical tree diameter growth with ontogenetic stage. *Oecologia* **175**: 1315–1324. doi:10.1007/s00442-014-2981-z.
- Soil Survey Staff. 1999. *Soil taxonomy: A basic system of soil classification for making and interpreting soil surveys*. 2nd edition. Natural Resources Conservation Service. U.S. Department of Agriculture Handbook 436.
- Su Y-S, Yajima M. 2015. *R2jags: a package for running jags from R*. R package version 0.05-01 **34**.
- Umaña MN, Zipkin EF, Zhang C, Cao M, Lin L, Swenson NG. 2018. Individual-level trait variation and negative density dependence affect growth in tropical tree seedlings. *Journal of Ecology* **106**: 2446–2455. doi:10.1111/1365-2745.13001.
- Uriarte M, Canham CD, Thompson J, Zimmerman JK. 2004. A neighborhood analysis of tree growth and survival in a hurricane-driven tropical forest. *Ecological Monographs* **74**: 591–614. doi:10.1890/03-4031.
- Violle C, Navas M-L, Vile D, et al. 2007. Let the concept of trait be functional!. *Oikos* **116**: 882–892. doi:10.1111/j.0030-1299.2007.15559.x.
- Wang XG, Wiegand T, Hao ZQ, Li BH, Ye J, Lin F. 2010. Species associations in an old-growth temperate forest in north-eastern China. *Journal of Ecology* **98**: 674–686. doi:10.1111/j.1365-2745.2010.01644.x.
- Wang XG, Comita LS, Hao ZQ, et al. 2012. Local-scale drivers of tree survival in a temperate forest. *PLoS One* **7**: e29469. doi:10.1371/journal.pone.0029469.
- Warton DI, Duursma RA, Falster DS, Taskinen S. 2012. smatr 3 – an R package for estimation and inference about allometric lines. *Methods in Ecology and Evolution* **3**: 257–259.
- Webb CO, Peart DR. 2000. Habitat associations of trees and seedlings in a Bornean rain forest. *Journal of Ecology* **88**: 464–478. doi:10.1046/j.1365-2745.2000.00462.x.
- Weemstra M, Mommer L, Visser EJW, et al. 2016. Towards a multidimensional root trait framework: a tree root review. *New Phytologist* **211**: 1159–1169. doi:10.1111/nph.14003.
- Westoby M. 1998. A leaf-height-seed (LHS) plant ecology strategy scheme. *Plant and Soil* **199**: 213–227.
- Westoby M, Wright IJ. 2006. Land-plant ecology on the basis of functional traits. *Trends in Ecology & Evolution* **21**: 261–268.
- Westoby M, Falster DS, Moles AT, Vesk PA, Wright IJ. 2002. Plant ecological strategies: some leading dimensions of variation between species.

- Annual Review of Ecology and Systematics* **33**: 125–159. doi:[10.1146/annurev.ecolsys.33.010802.150452](https://doi.org/10.1146/annurev.ecolsys.33.010802.150452).
- Wright IJ, Reich PB, Westoby M, *et al.* 2004. The worldwide leaf economics spectrum. *Nature* **428**: 821–827. doi:[10.1038/nature02403](https://doi.org/10.1038/nature02403).
- Wright SJ, Kitajima K, Kraft NJB, *et al.* 2010. Functional traits and the growth–mortality trade-off in tropical trees. *Ecology* **91**: 3664–3674. doi:[10.1890/09-2335.1](https://doi.org/10.1890/09-2335.1).
- Wright SJ, Yavitt JB, Wurzbarger N, *et al.* 2011. Potassium, phosphorus, or nitrogen limit root allocation, tree growth, or litter production in a lowland tropical forest. *Ecology* **92**: 1616–1625. doi:[10.1890/10-1558.1](https://doi.org/10.1890/10-1558.1).
- Wright SJ, Turner BL, Yavitt JB, *et al.* 2018. Plant responses to fertilization experiments in lowland, species-rich, tropical forests. *Ecology* **99**: 1129–1138. doi:[10.1002/ecy.2193](https://doi.org/10.1002/ecy.2193).
- Yang H, Wu Y. 1988. Tree composition, age structure and regeneration strategies of the mixed broad-leaf Korean pine forest in Changbaishan biosphere reserve, China. *Scientia Silvae Sinicae* **24**: 18–27.
- Yang H, Li D, Wang B, Han J. 1985. Distribution patterns of dominant tree species on northern slope of Changbai Mountain. *Forest Ecosystem Research* **5**: 1–14.
- Yang J, Cao M, Swenson NG. 2018. Why functional traits do not predict tree demographic rates. *Trends in Ecology & Evolution* **33**: 326–336. doi:[10.1016/j.tree.2018.03.003](https://doi.org/10.1016/j.tree.2018.03.003).
- Zemunik G, Davies SJ, Turner BL. 2018. Soil drivers of local-scale tree growth in a lowland tropical forest. *Ecology* **99**: 2844–2852. doi:[10.1002/ecy.2532](https://doi.org/10.1002/ecy.2532).
- Zhang ZC, Papaik MJ, Wang XG, *et al.* 2016. The effect of tree size, neighborhood competition and environment on tree growth in an old-growth temperate forest. *Journal of Plant Ecology* **10**: 970–980.
- Zhao Q, Zeng DH. 2019. Nitrogen addition effects on tree growth and soil properties mediated by soil phosphorus availability and tree species identity. *Forest Ecology and Management* **449**: 117478. doi:[10.1016/j.foreco.2019.117478](https://doi.org/10.1016/j.foreco.2019.117478).
- Zhu Y, Hogan JA, Cai HY, Xun YH, Jiang F, Jin GZ. 2017. Biotic and abiotic drivers of the tree growth and mortality trade-off in an old-growth temperate forest. *Forest Ecology and Management* **404**: 354–360. doi:[10.1016/j.foreco.2017.09.004](https://doi.org/10.1016/j.foreco.2017.09.004).
- Zhuang CY, Huang QL, Ma ZB, Luo F, Zhang Y. 2014. Review on defining methods for canopy stratification. *World Forestry Research* **27**: 34–40.